Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO2.2a	B1	$F - R = 3$
			Total 1 mark

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms two equations of motion for 1.8 and 1.2 kg mass, three terms, condone sign error. “Whole string” approach scores M0	AO3.4	M1	$1.8 \times 9.81 - T = 1.8 \times a$
	Obtains two correct equations	AO1.1b	A1	$T - 1.2 \times 9.81 = 1.2 \times a$
	Solves to find correct value for a $0.2g$ or AWRT 1.96	AO1.1b	A1	$a = 1.96$ m s^{-2}
	Uses $s = ut + \frac{1}{2}at^2$ with *their* calculated a value	AO1.1a	M1	$1.5 = \frac{1}{2} \times 1.96 \times t^2$
	Calculates correct t value AWRT 1.24	AO1.1b	A1	$t = 1.24$ seconds
(b)	Any valid assumption stated. Do not allow any comment already stated as an assumption in the question. Air resistance is permitted.	AO3.5b	E1	The string is long enough so that lighter mass does not reach the peg before the heavier mass hits the ground.
				Total 6 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States that the resistance is less for the second rider	AO2.4	E1	Jason experiences less air

	so the force required for equilibrium is also less, or mention of slipstreaming OE			resistance than Laura.
(b) (i)	Models situation by using given mass and total resistance force to form equation of motion PI	AO3.3	M1	As $F = ma$ then $a = -0.625$ m s^{-2}
	Uses appropriate suvat formula. May include $u = 25$.	AO3.4	M1	$u = 25$ km/h $= 6.944$ m s^{-1} $v = 0$; use $v^2 = u^2 + 2as$ so that $0 = 6.944^2 + 2 \times (-0.625) \times s$
	States correct value for s (AWFW 38.5 to 38.6) **Or** maximum $u = 7.07$ m s^{-1} or 25.5 km/h (AWRT) **Or** a needs to be < -0.603 (AWRT) **Or** Resistance needs to be > 38.5 N (AWFW 38.4 to 38.6) **Or** v^2 is -1.8 (AWRT) when $s = 40$	AO1.1b	A1	$s = 38.6$ m
	Makes appropriate comparison to conclude that Laura stops in time. Not necessary to see $38.6 < 40$, but comparison for other variables must be clear.	AO3.2a	E1	So Laura stops before reaching the accident
(ii)	States an assumption that, if incorrect, would **contradict** the conclusion in (i). (eg reaction time, diminishing resistive force as speed drops OE)	AO3.5a	E1F	Taking account of reaction time would mean she travelled a distance before starting to brake.
				Total 6 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses model for maximum friction $= \mu mg$	AO3.3	B1	$F_{max} = \mu mg$ $= 0.85 \times 20 \times 9.8$

	Marking Instructions	AO	Marks	Typical Solution
	Makes an appropriate comparison	AO1.1a	M1	= 166.6 N
	Explains clearly why crate remains stationary	AO2.4	E1	150 < 166.6 ∴ crate does not move
(b)	Forms an equation by resolving vertically Condone one of sign error or cos error	AO3.1b	M1	$20g = R + 150\sin 15°$ $R = 157.177$ N
	Obtains correct reaction force	AO1.1b	A1	$F_{max} = \mu \times 157.177$ $= 133.6$ N
	Uses maximum friction = μR With 'their' reaction force Must identify maximum or limiting friction	AO1.1b	B1F	150cos15° = 145 N 145 > 133.6 ∴ crate begins to move
	Compares 150cos15° with 'their' maximum friction	AO1.1a	M1	
	Explains, using their values, why the crate begins to move.	AO2.4	E1F	
				Total 8 marks

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Forms equation of motion with four correct terms Condone sign error	AO3.4	M1	$300 - 140 - R = 482 \times 0.2$ $R = 63.6$ N
	Obtains correct equation.	AO1.1b	A1	
	Obtains correct value of R.	AO1.1b	A1	
(ii)	Forms equation of motion with correct terms Condone sign error	AO1.1a	M1	$T - 63.6 = 72 \times 0.2$ $T = 78$ N
	Obtains correct equation Follow through their R	AO1.1b	A1F	
	Obtains correct value of T	AO1.1b	A1	
(b)	States appropriate assumption **NOT friction or air resistance**	AO3.3	E1	Rope has no mass **or** is horizontal **or** is inextensible

(c)(i)	Forms equation of motion for skater using 'their' R Condone sign error	AO3.1b	M1	$-63.6 = 72a$
	Finds correct acceleration for 'their' R	AO1.1b	A1F	$a = -0.883\ldots \text{ m s}^{-2}$
	Uses a suitable constant acceleration formula with 'their' a	AO1.1a	M1	$u = 6$ $v = 0$ $a = -0.883$
	Obtains s when $v = 0$ Or Obtains v or positive v^2 when $s = 20$	AO1.1b	A1F	$0 = 6^2 - 2 \times 0.883s$ $s = 20.4$ m
	Explains that the skater hits buggy using correct values	AO3.2a	E1	$20.4 > 20$ Skater hits buggy
(ii)	Explains that the tension is removed from the buggy	AO2.4	E1	The rope is released so there is no tension acting on the buggy, so there is a higher resultant force. The driver will notice an increase in acceleration.
	Explains that the driver notices an increase in acceleration	AO2.4	E1	
				Total 14 marks

Q6.

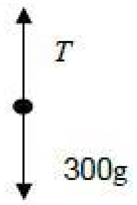
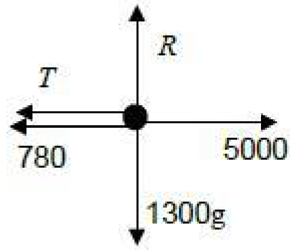
Marking Instructions	**AO**	**Marks**	**Typical Solution**
Applies Newton's 2nd Law to form a 3 term equation Award mark even if signs not correct	AO1.1a	M1	$F - 80 \times 10 = -80 \times 1.5$
Obtains a correct 3 term equation.	AO1.1b	A1	$F - 800 = -120$
Obtains correct reaction force. Must be given to 1 sf FT from incorrect 3 term equation provided M1 mark was awarded (condone omission of units)	AO1.1b	A1F	$F = 680 = 700$(N) to 1 sf
			Total 3 marks

Q7.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Differentiates, with at least one term correct	AO1.1a	M1	$\frac{dv}{dt} = 12t - 36^2$
	Selects and applies $F = ma$ to 'their' derivative Condone use of 400 for mass	AO1.1a	M1	$F = ma = 0.4\ (12t - 36t^2)$
	Obtains correct expression for force FT from 'their' $F = ma$ equation, provided the first M1 has been awarded (may be in factorised form)	AO1.1b	A1F	$= 4.8t - 14.4t^2$
(b)	Integrates v to find r, with at least one term correct	AO3.1b	M1	$r = \int (6t^2 - 12t^3)\, dt$
	Obtains correct integral (condone absence of c)	AO1.1b	A1	$r = 2t^3 - 3t^4 + c$
	Deduces the value of c using initial conditions FT use of 'their' integral provided M1 awarded	AO2.2a	A1F	When $t = 0$, $r = 0$ so $0 = 2 \times 0^2 - 3 \times 0^4 + c$ so $c = 0$
	Forms and solves an equation for t (condone numerical slip)	AO1.1a	M1	$0 = 2t^3 - 3t^4$ $= t^3(2 - 3t)$ $t = 0$ or $t = \frac{2}{3}$
	Interprets solution, realising that the non-zero time is required (Must include units) FT use of 'their' equation for t provided both M1 marks have been awarded	AO3.2a	A1F	Next at O at $\frac{2}{3}$ seconds
				Total 8 marks

Q8.

Marking Instructions	AO	Marks	Typical Solution

(a)(i)	Draws correct force diagram for crate from information given to use as a model in this context Must introduce a variable to represent the tension in the string	AO3.3	B1	
(ii)	Draws correct force diagram for van from information given to use as a model in this context Must introduce a variable to represent the tension in the string	AO3.3	B1	
(b)	Applies Newton's 2nd Law ($F = ma$) to the crate	AO3.4	M1	For crate $T - 300\text{g} = 300a$
	Applies Newton's 2nd Law ($F = ma$) to the van ($F = ma$ 'round the corner' scores 0)	AO3.4	M1	For van $5000 - T - 780 = 1300a$ $4220 - T = 1300a$
	Solve their simultaneous equations	AO1.1a	M1	$4220 - 300\text{g} = 1600a$
	Finds the value of a correctly **AG**	AO1.1b	A1	$a = 1280 \div 1600 = 0.80$ m s^{-2} **(AG)**
(c)	Uses $a = 0.80$ in either of their two equations in **(b)**	AO3.4	M1	$T = 300 \times 0.80 + 300\text{g}$
	Finds the correct value for T (condone omission of units) Possibly done in **(b)**	AO1.1b	A1	= 3180 = 3200 (N) (2 sf)
(d)	Explains that the model could be refined by including air resistance	AO3.5c	E1	Resistance will increase with speed
				Total 9 marks

Q9.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	80 m s^{-2}

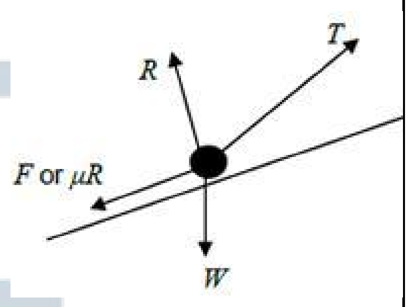
Total 1 mark

Q10.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates two (or four) appropriate distances	AO3.1b	M1	$s_1 = \frac{1}{2}(6 + 10) \times 8 = 64$ m
	Obtains correct distances	AO1.1b	A1	$s_2 = \frac{1}{2} \times 10 \times 2 = 10$ m
	Obtains correct sum of 'their' distances	AO1.1b	A1F	$s_1 + s_2 = 64 + 10 = 74$ m **OR** $S_1 = 6 \times 8 = 48$ m $S_2 = \frac{1}{2} \times 4 \times 8 = 16$ m $S_3 = \frac{1}{2} \times 4 \times 2 = 4$ m $S_4 = \frac{1}{2} \times 6 \times 2 = 6$ m $S_1 + S_2 + S_3 + S_4 = 74$ m
(b)	Finds difference of 'their' distances from part **(a)**	AO2.2a	B1F	$64 - 10 = 54$ m
(c)	Calculates magnitude of acceleration	AO3.1b	M1	$a_{max} = \frac{8}{4} = 2$
	Obtains correct resultant force	AO1.1b	A1	$F_{max} = 800 \times 2 = 1600$ N
(d)	Explains that abrupt changes and straight lines in the graph are unlikely in reality	AO3.5b	E1	Change of velocity is unlikely to result in abrupt changes I would expect to see curves on the graph
				Total 7 marks

Q11.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Resolves horizontally and vertically to obtain expressions for F and R	AO3.4	M1	Resolving vertically $R = 8 \times 9.8 - 50\sin 40°$ Resolving horizontally

	(allow consistent mixing of sin and cos)			$F = 50\cos 40°$
	Obtains correct expressions for R and F	AO1.1b	A1	
	States friction model $F = \mu R$ with 'their' values for F and R	AO1.2	B1	$F = \mu R$ $50\cos40° = \mu(8 \times 9.8 - 50\sin40°)$ $\mu = \dfrac{50\cos 40°}{8 \times 9.8 - 50\sin 40°}$
	Completes a rigorous argument that results with correct μ **AG** Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$\mu = 0.8279660445 = 0.83$ (2sf) **AG**
(b) (i)	Draws correct diagram with exactly four forces showing arrow heads and labels Can use Mg or $8g$ or 78.4 for W	AO3.3	B1	
(ii)	Resolves perpendicular to the plane resulting in a three term equation containing; R, $8g\cos 5°$ and $T\sin 40°$ (or $T\cos 50°$) OR Resolves horizontally and vertically to obtain equations of motion in horizontal and vertical directions	AO3.1b	M1	$R = 8 \times 9.8\cos 5° - T\sin 40°$ $= 78.1 - T\sin 40°$
	Obtains correct expression for R OR Obtains correct horizontal and vertical equations	AO1.1b	A1	$T\cos 40° - 8 \times 9.8\sin 5° - F = 8 \times 3$ $T\cos 40° - 8 \times 9.8\sin5° - 0.827\ldots \times (8 \times 9.8\cos 5° - T\sin 40°) = 24$
	Forms a four term equation of motion parallel to the plane with correct terms	AO3.1b	M1	$T = \dfrac{24 + 8 \times 9.8\sin5° + 0.827\ldots \times 8 \times 9.8\cos5°}{\cos40° + 0.827\ldots \times \sin40°}$ $T = 73.55939193 = 74$ (2

(allow sign errors) OR Eliminates R to solve for T			*sf)*
Obtains correct equation of motion OR Obtains correct expression(s) without R	AO1.1b	A1	
Uses the friction model in the four term equation of motion where R is in the form $a - bT$ and a and b are positive constants OR Uses the friction model in the horizontal and vertical equations Dependent on previous M1	AO3.1b	dM1	
Solves for T	AO1.1b	A1	
Obtains correct value of T with 2 sf accuracy. FT incorrect value found for T provided both M1 marks and dM1 mark have been awarded	AO3.2a	A1F	
			Total 12 marks

Q12.

(a) $(F =) 9\mathbf{i} - 3\mathbf{j} + 5\mathbf{i} + 8\mathbf{j} - 7\mathbf{i} + 3\mathbf{j} = 7\mathbf{i} + 8\mathbf{j}$

M1: Adding the three forces with one component correct.

A1: Correct answer.

M1A1

2

(b) $(F =) \sqrt{7^2 + 8^2} = \sqrt{113} = 10.6$ N

M1: Finding magnitude with a + sign.

A1F: Correct magnitude. Accept AWRT 10.63 and $\sqrt{113}$

Follow through incorrect answers to part (a).

M1A1F

2

(c) $(a =) \dfrac{\sqrt{113}}{5} = 2.13 \text{ m s}^{-2}$

M1: Dividing their force from part (a) or magnitude by 5.
A1F: Correct acceleration.

Accept 2.12 (from truncation or 10.6 / 5) or $\frac{\sqrt{113}}{5}$ *or AWRT 2.13.*
Follow through incorrect answers to parts (a) and (b).
*Seeing just **a** = 1.4**i** + 1.6**j** scores M1 A0*

M1A1F

2

(d) $\cos \alpha = \frac{7}{\sqrt{113}}$ or $\frac{7}{10.6}$

M1: Trig equation to find the angle with: cos with 7 or 8 in the numerator and $\sqrt{113}$ *in denominator*

OR

$\sin \alpha = \frac{8}{\sqrt{113}}$ or $\frac{8}{10.6}$

sin with 7 or 8 in the numerator and $\sqrt{113}$ *in denominator*

OR

$\tan \alpha = \frac{8}{7}$

tan with 7 and 8 in any position
A1F: Correct equation.

M1A1F

(α =)48.8°

A1F: Correct angle. Accept 49° or AWRT 49°
Follow through incorrect answers to parts (a) and (b).

A1F

3

[9]

Q13.

(a)

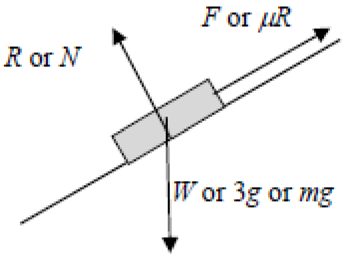


Diagram with exactly three forces showing arrow heads and *labelled. If components are also shown they must use a different style e.g. dashed lines then they can be ignored. Friction must be up the slope.*

B1

1

(b) (R =)3 × 9.8 cos 40° = 22.5 N

M1: Resolving perpendicular to the slope. Can use sin40° or cos50° for method mark, with g or 9.8.

A1: Correct normal reaction. Accept AWRT 22.5 (Note use of 9.81 still gives 22.5 N.)

M1A1

2

(c) (F =)0.2R = 4.50 N

M1: Use of $F = \mu R$.

A1F: Correct friction. Accept 4.5 N or AWRT 4.50.

(Accept 4.51 N from the use of 9.81.)

Follow through incorrect normal reaction from part (b).

M1A1F

2

(d) 3a = 3 × 9.8sin 40° – 4.504

M1: Three term equation of motion with correct terms, with 3a, either component of weight and their answer to part (c) for F.

A1F: Equation of motion with correct terms and signs.

M1A1F

a = 4.80 m s^{-2}

A1F: Correct acceleration. Accept 4.8 or AWRT 4.80.

(Note that using 9.81 still gives 4.80 m s^{-2}).

Follow through friction from part (c).

A1F

3

(e) No air resistance force acting
or
No other forces acting on the box.
or
They (forces in the diagram) are the only forces that act.
OR
No turning effect (due to forces).
or
Forces are concurrent. OE

B1: Correct assumption.

Ignore irrelevant comments

B1

1

[9]

Q14.

(a) 5900 × 0.2 = 2500 – 800 – R

M1: Equation of motion for tractor and trailer as a single particle, with 2500, 800, R (which might be implied by seeing 1180 and 1700 or 1180 and 3300) and 5900 × 0.2 OE, with any signs.

A1: Correct equation.

M1A1

(R =)2500 – 1180 – 800 = 520 N

A1: Correct R.

A1

If tension found first, do not award any marks until an equation for R is obtained. Award M1 for 3500 × 0.2 = ±2500 ± R ± 1280.

3

(b) T – 800 = 2400 × 0.2

M1: Equation for trailer with 2400 and 800.

A1: Correct equation.

M1A1

(T =)800 + 480 = 1280 N

A1: Correct tension.

A1

3

OR

3500 × 0.2 = 2500 – 520 – T

M1: Equation for tractor with 3500, 2500 and 520.

A1F: Correct equation.

(M1A1F)

(T =)2500 – 700 – 520 = 1280 N

A1F: Correct tension.

(A1F)

(3)

Follow through incorrect R from part (a).

If the tension has been found in part (a) it only needs to be stated here.

(c) 1280 N

B1F: Same answer as part (b).
Do not accept – 1280.

B1F

1

[7]

Q15.

(a) Using $F = ma$:

$$-4v^{\frac{1}{3}} = 12\frac{\mathrm{d}v}{\mathrm{d}t}$$

B1

$$\therefore \frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{1}{3}v^{\frac{1}{3}}$$

$$-3\int \frac{\mathrm{d}v}{v^{\frac{1}{3}}} = \int \mathrm{d}t$$

condone –, 3 incorrect side

M1

$$-3 \times \frac{v^{\frac{2}{3}}}{\frac{2}{3}} = t + c$$

$$-\frac{9}{2}v^{\frac{2}{3}} = t + c$$

condone lack of + c

A1

When $t = 0$, $v = 8 \Rightarrow c = -18$

M1A1

$$-\frac{9}{2}v^{\frac{2}{3}} = t + 18$$

$$v^{\frac{2}{3}} = 4 - \frac{2}{9}t$$

$v = \left(4 - \frac{2}{9}t\right)^{\frac{3}{2}}$

A1

6

(b) Particle is at rest when $4 - \frac{2}{9}t = 0$

The value of t is 18

B1

1

[7]

Q16.

(a) $3g - T = 3a$

M1: Three term equation of motion with $3g$ or 29.4, T and $3a$.
A1: Correct equation.

M1A1

$T - g = a$

$2g = 4a$

M1: Three term equation of motion with g or 9.8, T and a.
A1: Correct equation.

M1A1

$a = \frac{g}{2} = 4.9 \text{ m s}^{-2}$

A1: Correct final answer. Accept $\frac{g}{2}$

A1

Note: Do not penalise candidates who consistently use signs in the opposite direction throughout, provided they then give their final answer as 4.9, having seen –4.9 in their working. If the final answer is –4.9 don't award the final A1 mark.
Special Case:

Whole string method $2g = 4a$ and $a = \frac{2g}{4} = 4.9$ OE scores M1A1A1

5

(b) $v^2 = 0^2 + 2 \times 4.9 \times 0.4$

M1: Use of a constant acceleration equation to find v, with u = 0, their value for a from part (a) and s = 0.4 or 40.

M1

$v = \sqrt{3.92} = 1.98 \text{ m s}^{-1}$

A1F: Correct speed. Follow through their acceleration from part (a).
Use $v = \sqrt{0.8a}$ for FT.
Accept 1.97.
If candidates use two equations, award no marks until they have an equation for v. (Note use of t = 0.404 or better required for A1)

A1F

2

(c) $0^2 = \left(\sqrt{3.92}\right)^2 + 2 \times (-9.8)\, s$

M1: Use of a constant acceleration equation with v = 0, a = ±9.8 and their speed from (b).

M1

$s = \dfrac{3.92}{2 \times 9.8} = 0.2 \text{ m}$

A1: Correct distance.

A1

Total = 0.2 + 0.4 = 0.6 m

A1: Correct total distance. Allow 60 cm from correct working.
Note
$0^2 = \left(\sqrt{392}\right)^2 + 2 \times (-9.8)s$
$s = \dfrac{392}{2 \times 9.8} = 20$
scores M1A0A0

A1

If candidates use two equations, award no marks until they have an equation for s. (Note use of t = 0.202 or better required for A marks)

3

(d) The acceleration would be less, because the resultant force on each particle would be reduced.

B1: Less
'Slower acceleration' not acceptable

B1

B1: Appropriate reason.
Only award second B1 if they say acceleration is less.

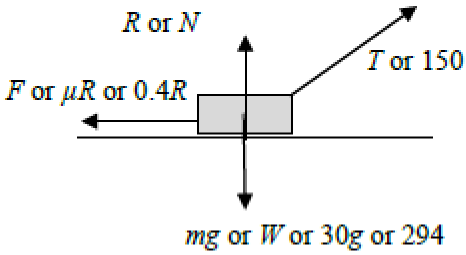
B1

2

[12]

Q17.

(a) (i)



B2: Correct diagram with exactly four forces showing arrow heads and labelled.
B1: Diagram with one error or omission.
B0: Diagram with 2 or more errors or omissions.

If components are also shown and they use a different style, eg dashed lines, they can be ignored.

If both components are shown in the same style as other forces, this counts as two errors.

Note; Do not accept 30kg for the weight.

B1

B1

2

(ii) $R + 150 \sin 20° = 30 \times 9.8$

$(R =)\ 30 \times 9.8 - 150 \sin 20°$

$= 242.69\ldots\ldots$

M1: Resolving vertically to obtain a three term equation, with R, 150 sin or cos(20° or 70°) and 30g oe.

A1: Correct equation. Allow g instead of 9.8.

M1A1

= 243 N (to 3sf)

A1: AG Correct final answer having seen either 2nd or 3rd or both line of solution.

A1

3

(iii) $(F =)\ 0.4 \times 242.7 = 97.1$ N

M1: Use of $F = \mu R$ or $F \le \mu R$.
A1: Correct final answer without an inequality.
Accept 97.2.

M1A1

2

(iv) $30a = 150 \cos 20° - 97.08$

M1: Three term equation of motion with 30a, 150 sin or cos(20 or 70°) and their friction from (a)(iii). Condone incorrect signs.
A1: Correct equation.

M1A1

$$a = \frac{150\cos 20° - 97.08}{30} = 1.46 \text{ m s}^{-2}$$

dM1: Solving for a.

dM1

A1: Correct acceleration. Accept 1.45 or 1.47 or AWRT 1.46.

A1

4

(b) $R = 30 \times 9.8 - T\sin 20°$

B1: Correct normal reaction in terms of T.

B1

$F = 0.4\,(30 \times 9.8 - T\sin 20°)$

B1: Correct friction in terms of T

B1

$T\cos 20° = 0.4\,(30 \times 9.8 - T\sin 20°)$

M1: Resolving tension horizontally and equating to F, provided that F is in terms of T.
A1: Correct equation.

M1A1

$$T = \frac{0.4 \times 30 \times 9.8}{\cos 20° + 0.4\sin 20°} = 109 \text{ N}$$

A1: Correct tension. AWRT 109.

A1

5

(c) The same

B1: The same.

Use of g = 9.81 gives acceptable final answers.

B1

1

[17]

Q18.

(a) $v = \frac{ds}{dt}$

M1

$= 24t^2$

A1

2

(b) $a = \frac{dv}{dt}$

$= 48t$

B1

When $t = 2$, $a = 96$

B1

Using $F = ma$

$F = 3 \times 96$

M1

$= 288$ N

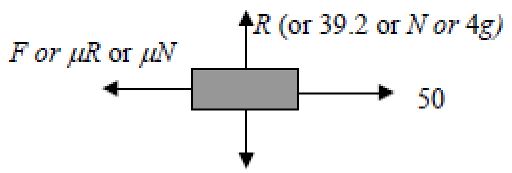
A1

4

[6]

Q19.

(a)



mg or W or $4g$ or 39.2

B1: Correct force diagram with four forces with arrows and labels. Accept words eg friction instead of letters.
Ignore negative signs in labels.
Do not accept 4 kg for the weight.
Award marks if forces are drawn on the diagram in the question.

B1

1

(b) 39.2 N

B1: Correct reaction force.
Accept 4g. Do not accept 39.

B1

1

(c) $50 - F = 4 \times 3$

M1: Three term equation of motion with the correct terms.
A1: Correct equation with correct signs.

M1A1

$F = 38$

A1: Correct friction.

A1

3

(d) $38 = \mu \times 39.2$

M1: Use of $F = \mu R$ with their answers to (b) and (c).

M1

$\mu = \frac{38}{39.2} = 0.969$

A1F: Correct μ based on their answers to (b) and (c).
Accept AWRT 0.969.

A1F

Note: F = 12 leads to 0.306 and award M1 A1F

Condone 0.97 or FT to 2sf

Condone use of inequalities.

2

(e) Less friction, so a smaller coefficient of friction.

B1: Less friction.

B1

B1: Smaller μ.

B1

Note:

More friction anywhere scores B0 B0

Less friction, greater μ scores B1 B0

Smaller μ with no / inexact reason B0 B1

2

[9]

Q20.

(a) $4720 - 3R = 2200 \times 1.6$

M1: Three term horizontal equation of motion with mass of 2200 kg and $3R$ (or $2R$ and R).

A1: All terms correct (4720, $3R$ and 2200×1.6).

M1A1

A1: Correct signs.

A1

$R = \dfrac{4720 - 3520}{3} = 400$

A1: Correct value for R.

A1

OR

$4720 - R - T = 1200 \times 1.6$

$T - 2R = 1000 \times 1.6$

$4720 - 3R = 3520$

M1: Forming an equation for each body and adding to eliminate T.

A1: Two correct equations.

(M1A1)

A1: Correct equation in R.

(A1)

$R = 400$

A1: Correct value for R.

(A1)

4

(b) $T - 2 \times 400 = 1000 \times 1.6$

M1: Three term equation of motion for caravan with T,

$2R$ and 1000 × 1.6.

A1F: Correct equation, with their value for R from part (a).

M1A1F

T = 800 + 1600 = 2400 N

A1F: Correct tension. Follow through from part (a) using T = 1600 + $2R$

A1F

OR

4720 – T – 400 = 1200 × 1.6

M1: Four term equation of motion for car with 4720, T, R and 1200 × 1.6.

(M1)

T = 4720 – 400 – 1920 = 2400 N

A1F: Correct equation, with their value for R from part (a)

(A1F)

A1F: Correct tension. Follow through from part (a) using T = 2800 – R

Note: do not follow through if a negative value is used for R.

(A1F)

3

[7]

Q21.

(a) (i) $10^2 = 4^2 + 2 \times a \times 50$

M1: Use of a constant acceleration equation to find a, with v and u substituted correctly.

For example $4^2=10^2 + 100a$ scores M0A0A0.

A1: Correct constant acceleration equation.

M1A1

$$a = \frac{100-16}{100} = 0.84 \text{ ms}^{-2}$$

A1: Correct a.

A1

Note if t found first award M1 for use of

$v = u + at$ *or* $s = ut + \frac{1}{2}at^2$.

3

(ii) $50 = \frac{1}{2}(4 + 10)t$

M1: Use of a constant acceleration equation to find t.

A1F: Correct constant acceleration equation with their acceleration from (a)(i) seen.

M1A1

$t = \frac{50}{7} = 7.14$ s

A1: Correct t. Accept $\frac{50}{7}$ or $7\frac{1}{7}$ or AWRT 7.14.

A1

OR

$10 = 4 + 0.84t$

If t has been found in part (a) the working does not have to be repeated, but value of t must be stated.

(M1A1F)

$t = \frac{6}{0.84} = 7.14$ s

Do not follow through incorrect values of a.

(A1)

OR

$50 = 4t + \frac{1}{2} \times 0.84t^2$

(M1A1F)

$0.42t^2 + 4t – 50 = 0$

$t = 7.14$ (or $t = –16.6$)

(A1)

3

(b) $70 \times 0.84 = 58.8$ N

M1: Use of $F = ma$ with $m = 70$ and their a from (a)(i).

A1F: Correct F. Follow through their value of a from part (a)(i).

M1A1F

2

(c) (i) $58.8 = 70 \times 9.8 \sin \alpha$

$\sin \alpha = \frac{58.8}{70 \times 9.8} = 0.08571$

M1: Resolving parallel to the slope must see 70g or mg OE with sin α or cos α and their answer to part (b).

A1F: Correct equation. Follow through their answer to part (b) provided sin α < 1

M1A1F

$\alpha = 4.92°$

A1F: Correct angle. Follow through their answer to part (b). Accept 4.91° provided sin α < 1.

A1F

3

(ii) $70 \times 9.8 \sin \alpha - 30 = 58.8$

$\sin \alpha = 0.12945$

M1: Three term equation of motion. Must see 70g or mg OE with sin α or cos α.

A1F: Correct equation. Follow through their answer to part (b) provided sin α < 1

M1A1F

$\alpha = 7.44°$

A1F: Correct angle. Follow through their answer to part (b) provided sin α < 1.

Accept 7.43°.

Accept 7.41° from 0.129.

A1F

3

(d) The air resistance force will increase (vary or change) with speed.

B1: Correct statement.

B1

1

[15]

Q22.

(a) using $\mathbf{F} = m\mathbf{a}$:

ie dividing by 50

M1

$\mathbf{a} = (6t - 1.2\,t^2)\,\mathbf{i} + 2\,e^{-2t}\,\mathbf{j}$

A1

2

(b) $\mathbf{v} = \int \mathbf{a}\,dt$

$= (3\,t^2 - 0.4\,t^3)\,\mathbf{i} - e^{-2t}\mathbf{j} + \mathbf{c}$

when $t = 0$, $\mathbf{r} = 7\,\mathbf{i} - 4\,\mathbf{j}$

condone lack of + c; M1 one term correct

M1A1

$\mathbf{c} = 7\,\mathbf{i} - 3\mathbf{j}$

$\mathbf{v} = (7 + 3\,t^2 - 0.4\,t^3)\,\mathbf{i} - (3 + e^{-2t})\,\mathbf{j}$

*ft from ke^{-2t} in (b); just adding 7**i** – 4**j**, m0 accept unsimplified. CAO*

m1A1

4

(c) when $t = 1$, $\mathbf{v} = 9.6\,\mathbf{i} - 3.135\,\mathbf{j}$

ft from (b)

M1A1

speed = $\sqrt{9.6^2 + 3.135^2}$

m1

$= 10.1\ \text{ms}^{-1}$

ft from (b)

A1

4

[10]

Q23.

(a) $10^2 = 20^2 + 2 \times a \times 75$

M1: Use of a constant acceleration equation to find a, with $v = 10$ and $u = 20$.

$20^2 = 10^2 + 2 \times a \times 75$ scores M0

A1: Correct equation.

M1A1

$a = \frac{100 - 400}{150} = -2 \text{ ms}^{-2}$

A1: Correct acceleration.
For two equation methods award no marks until an equation for a is obtained.

A1

3

(b) $0 = 20 - 2t$

M1: Using a constant acceleration equation, with u = 20 and v = 0, to find t using their acceleration from (a) even if positive.
Using s = 75 scores M0

M1

t = 10 seconds

A1: Correct time from correct working CSO.

A1

2

(c) $F = 1400 \times 2 = 2800$ N

M1: Use of $F = ma$ with ± their acceleration and mass of 1400.
A1F: Correct force. Follow through the magnitude of their acceleration. Answer must be positive. Sign changes do not need to be justified.

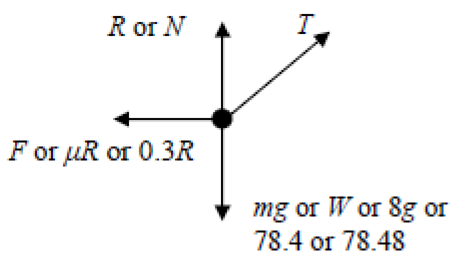
M1A1F

2

[7]

Q24.

(a)



*B1: Diagram with **exactly four** forces showing arrow heads and labelled. If components are also shown and they use a different style, eg dashed lines, they can be ignored.*
Note: Award mark if forces drawn on the diagram in the

question.

Note: Do not accept 8 kg for the weight.

Note Accept μR or 0.3R for F.

B1

1

(b) $R + 40 \sin \theta = 8 \times 9.8$

M1: Resolving vertically to obtain a three term equation, with R, T sin or cos(30° or 60°) and 8g oe.

A1: Correct equation

M1A1

$R = 78.4 - 40\sin \theta$

A1: Correct expression for R.

Accept (R=)8g – T sin30°

Note if using g = 9.81 accept

R = 78.48 – 0.5T or R = 78.5 – 0.5T

A1

3

(c) $8a = 40 \cos \theta - 0.3(78.4 - 40 \sin \theta)$

M1: Use of the friction inequality with R from part (b)

A1: Correct equation.

M1A1

$8a = -23.52 + 40 \cos \theta + 12 \sin \theta$

M1: Use of equation of motion with 40 sin θ or 40 cosθ and 8a and a friction term.

M1

$p = -2.94$

$q = 8$

$r = 1.5$

A1: One correct value.

A1

A1: All three values correct.

A1

5

[9]

Q25.

(a) (i) $10^2 = 20^2 + 2 \times a \times 75$

M1: Use of a constant acceleration equation to find a, with v = 10 and u = 20.
$20^2 = 10^2 + 2 \times a \times 75$ scores M0
A1: Correct equation.

M1A1

$a = \dfrac{100 - 400}{150} = -2\ \text{ms}^{-2}$

A1: Correct acceleration.

A1

For two equation methods award no marks until an equation for a is obtained.

3

(ii) $0 = 20 - 2t$

M1: Using a constant acceleration equation, with u = 20 and v = 0, to find t using their acceleration from (a) even if positive.
Using s = 75 scores M0

M1

t = 10 seconds

A1: Correct time from correct working CSO.

A1

2

(iii) $F = 1400 \times 2$

M1: Use of $F = ma$ with ± their acceleration and mass of 1400.

M1

= 2800 N

A1F: Correct force. Follow through the magnitude of their acceleration. Answer must be positive. Sign changes do not need to be justified.

A1F

2

(b) F = 2800 – 200 = 2600 N

B1F: The magnitude of their force minus 200.
Do not award if M1 not awarded in (a)(iii).
Final answer must be positive.
Follow through only if their answer to (a)(iii) is greater than 200.

B1F

1

[8]

Q26.

(a) $18g - T = 18a$

M1: Three term equation of motion for the 18 kg particle.
A1: Correct equation of motion for the 18 kg particle.
(Accept $T - 18g = 18a$)

M1A1

$T = 12a$

B1: Equation of motion for the block that has signs consistent with the first equation.

B1

$18g - 12a = 18a$

$a = \frac{18g}{30} = 5.88 \text{ ms}^{-2}$

A1: Correct acceleration from correct work. Accept $\frac{3g}{5}$
Do not penalise consistent use of negative acceleration, provided final answer positive.
Special Case:

Whole String Method $18g = 30a$ and $a = \frac{18g}{30} = 5.88$
OE M1A1
Note using $g = 9.81$ gives 5.89, also accept 5.88.

A1

4

(b) (i) $18g - T = 18 \times 3$

M1: Three term equation of motion for the 18 kg particle with $a = 3$ seen.
A1: Correct equation.

M1A1

$T = 18g - 18 \times 3 = 122(.4)\text{N}$

A1: Correct tension. Accept 122.4.
Note using $g = 9.81$ gives 123, also accept 122.

A1

3

(ii) (R = 12 × 9.8 = 117.6 N =)118 N(to 3sf)

B1: Correct normal reaction. Accept 117 and 117.6. Final answer must be positive.

Do not accept 12g.

Note using g = 9.81 gives 118, also accept 117.

B1

1

(iii) 122.4 – F = 12 × 3

M1: Three term equation of motion for the block, containing their tension, F and 12 × 3.

A1F: Correct equation. Follow through T from part (b)(i).

M1A1F

F = 86.4

A1F: Candidate's T minus 36.

A1F

86.4 = μ × 117.6

dM1: Use of $F = \mu R$ with AWRT 117 or 118 for R and the candidate's value of F provided positive.

dM1

$\mu = \frac{86.4}{117.6} = 0.735$

A1: Correct μ. Accept anything between 0.728 and 0.739 inclusive. Allow 0.73 and 0.74.

A1

5

(c) No air resistance or no other forces

B1: One assumption from list

B1

Horizontal String

Block is a particle or **they** are particles

B1: For another assumption from list.

Do not penalise assumptions not in the list.

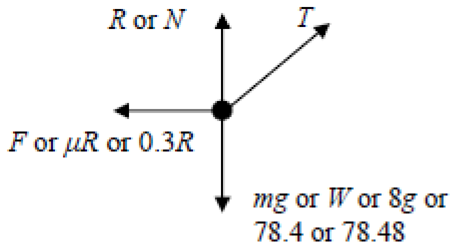
B1

2

[15]

Q27.

(a)



*B1: Diagram with **exactly four** forces showing arrow heads and labelled. If components are also shown and they use a different style, eg dashed lines, they can be ignored.*

Note: Award mark if forces drawn on the diagram in the question.

Note: Do not accept 8 kg for the weight.

Note Accept μR or 0.3R for F.

B1

1

(b) $R + T\sin 30° = 8 \times 9.8$

M1: Resolving vertically to obtain a three term equation, with R, T sin or cos(30° or 60°) and 8 g oe.

A1: Correct equation

M1A1

$(R =)\ 78.4 - T\sin 30°$

$(R =)\ 78.4 - 0.5\,T$

A1: Correct expression for R.

Accept $(R =)\ 8g - T\sin 30°$

Note if using g = 9.81 accept
$R = 78.48 - 0.5T$ or $R = 78.5 - 0.5T$

A1

3

(c) $T\cos 30° - F = 8 \times 0.05$

M1: Horizontal equation of motion with F, T sin or cos (30° or 60°) and 8×0.05 oe.

A1: Correct equation.

M1A1

$F = 0.3(78.4 - \sin 30°)$

M1: Using $F = 0.3R$ with their R from part (b), provided it includes a term in T.

A1: Correct expression for friction.

M1A1

$T\cos 30° - 0.3(78.4 - T\sin 30°) = 0.4$

$T = \dfrac{23.52 + 0.4}{\cos 30° + 0.3\sin 30°} = 23.5\text{N}$

dM1: Solving for T. Must see (cos 30° ± 0.3sin 30°) or similar in the denominator. (Dependent on both previous M marks.)

A1: Correct T. Accept 23.6 or AWRT 23.5

dM1A1

6

Or

$T\cos 30° - F = 8 \times 0.05$

M1: Horizontal equation of motion with F, T sin or cos(30° or 60°) and 8 × 0.05 oe.

A1: Correct equation.

(M1A1)

$T\cos 30° - 0.3R = 8 \times 0.05$

M1: Using $F = 0.3R$

A1: Two correct equations involving only T and R.

(M1A1)

$R + T\sin 30° = 8 \times 9.8$

solving simultaneously gives

$T = 23.5$

dM1: Solving for T.

A1: Correct T. Accept 23.6 or AWRT 23.5

Note using g = 9.81 gives 23.6, also accept 23.5.

(dM1A1)

[10]

Q28.

(a) (i) $a = \dfrac{dv}{dt}$

$= 12t + 8e^{-4t}$ m s^{-2}

M1 for either term correct

M1A1

2

(ii) When t = 0.5, $a = 6 + 8 \times e^{-2}$

m1

$= 7.08 \text{ m s}^{-2}$

Condone 7.07

SC1 for 7.1 with no working

A1

2

(b) Using $F = ma$:

$F = 4 \times 7.08$

$= 28.3$ N

Ft from value awarded A1

B1ft

1

(c) $r = \int v\,\mathrm{d}t$

At least two terms correct

M1

$= 2t^3 + \frac{1}{2}e^{-4t} + 8t + c$

Does not need + c

A1

When $t = 0$, $r = 0 \rightarrow c = -\frac{1}{2}$

Does not need $c = -\frac{1}{2}$

m1

$r = 2t^3 + \frac{1}{2}e^{-4t} + 8t - \frac{1}{2}$

Need r, s (or words)

A1

4

[9]

Q29.

(a) Using $F = ma$:

$m\frac{dv}{dt} = 49 - 9.8v$ or $5g$ 9.8 □v

Need to see $m\frac{dv}{dt}$ *or* $5\frac{dv}{dt}$ *or* $a = \frac{49-9.81}{5}$

M1

$\therefore \frac{dv}{dt} = -1.96(v-5)$

Must see m *terms (not* $a = \ldots$*)*

A1 **2**

(b) $\int \frac{dv}{v-5} = -1.96 \int dt$

And one side integrated

M1

$\ln(v-5) = -1.96t + c$

Need + c*, A1 each side*

A1A1

When $t = 0$, $v = 7 \Rightarrow c = \ln 2$

OE

A1

$\ln\frac{v-5}{2} = -1.96t$

$\frac{v-5}{2} = e^{-1.96t}$

$v = 5 + 2e^{-1.96t}$

CAO

A1 **5**

[7]

Q30.

(a) $s = \frac{1}{2} \times 10 \times 4 + 10 \times 4 + \frac{1}{2} \times (4+7) \times 10 + \frac{1}{2} \times 7 \times 10$

(= 20 + 40 + 55 + 35)

M1: Any one term correct.

M1: A second term correct.

A1: Correct expression for total distance.

M1M1A1

= 150 m

A1: Total distance correct.

A1

OR

$s = \frac{1}{2} \times (10 + 20) \times 4 + \frac{1}{2} \times (4 + 7) \times 10 + \frac{1}{2} \times 7 \times 10$

(= 60 + 55 + 35)

= 150 m

(M1M1A1)
(A1)

OR

$s = \frac{1}{2} \times 10 \times 4 + 10 \times 4 + 10 \times 4 + \frac{1}{2} \times 10 \times 3 + \frac{1}{2} \times 7 \times 10$

(= 20 + 40 + 40 + 15 + 35)

= 150 m

(M1M1A1)
(A1)

4

(b) Average Speed = $\frac{150}{40}$ = 3.75 m s^{-1}

M1: Their total distance divided by 40.

M1

A1F: Correct average speed based on their distance from part (a). Must be correct to three or more significant figures.

A1F

2

(c) $a = \frac{4}{10}$ = 0.4 m s^{-2}

M1: Any division involving the numbers 10 and 4.

M1

A1: Correct acceleration. CAO

A1

***Note on use of constant acceleration equations**:*
award M1 for correct equation with correct values and A1 for correct final answer.

2

(d) F = 200000 × 0.4 = 80000 N

M1: Multiplication of, 2 × 10^n, for any integer n, by candidate's acceleration from part (c).

A1F: Correct force based on their answer to part (c) multiplied by 200000.

M1A1F

Note: use of a = 2.5 gives 500000 N

Accept 80 kN

2

[10]

Q31.

(a) (i) P – 500 = 2200 × 0.8
P = 1760 + 500

M1: Equation of motion for car and caravan as a single body. Must see 2200 (or 1200 + 1000) multiplied by 0.8, and 500 (or 200 + 300). Allow sign errors.

A1: Correct equation.

M1A1

= 2260

A1: Correct value for P.

A1

(Award full marks for: (P =) 1760 + 500 = 2260 or similar to obtain correct final answer.)

OR (If finding the tension first)

P – 1100 – 200 = 1200 × 0.8
P = 960 + 1100 + 200

M1: Equation of motion for car with their value for the tension.
Must see 1200 multiplied by 0.8, 200 and their tension.

Allow sign errors.

A1: Correct equation.

(M1A1)

= 2260

A1: Correct value for P.

(A1)

(Award full marks for: (P =) 960 + 200 + 1100 = 2260 or similar to obtain correct final answer.)

3

(ii) $T - 300 = 1000 \times 0.8$

$T = 300 + 800$

M1: Equation of motion for caravan.

Must see 300 and 1000 multiplied by 0.8.

Allow sign errors.

A1: Correct equation.

M1A1

= 1100

A1: Correct tension. CAO

A1

OR

$2260 - 200 - T = 1200 \times 0.8$

$T = 2260 - 200 - 960$

M1: Equation of motion for car. Must see 2260 (or candidate's P), 200 and 1200 multiplied by 0.8. Allow sign errors.

A1: Correct equation.

(M1A1)

= 1100 N

A1: Correct tension. CAO

(A1)

If candidates find tension first it must be stated in part (a)(ii) to gain any marks.

The working does not have to be repeated if seen in part (a)(i).

3

(b) (i) $15 = 7 + 0.8t$

M1: Use of a constant acceleration equation to find t, with 7, 15 and 0.8.

A1: Correct equation.

M1A1

$t = \frac{15-7}{0.8}$ = 10 seconds

A1: Correct time. CAO

A1

3

(ii) $15^2 = 7^2 + 2 \times 0.8s$

M1: Use of a constant acceleration equation to find s, with 7, 15 and 0.8.

A1: Correct equation

M1A1

$s = \frac{15^2 - 7^2}{1.6} = 110$ m

A1: Correct distance. CAO

A1

OR

$s = \frac{1}{2}(7 + 15) \times 10 = 110$m

M1: Use of a constant acceleration equation to find s, with 7, 15 and candidate's time.

A1F: Correct equation.

A1F: Correct distance.

(M1A1F)
(A1F)

OR

$s = 7 \times 10 + \frac{1}{2} \times 0.8 \times 10^2 = 110$ m

M1: Use of a constant acceleration equation to find s, with 7, 0.8 and candidate's time.

A1F: Correct equation.

A1F: Correct distance.

(M1A1F)
(A1F)

If candidates find distance first it must be stated in part (b)(ii) to gain any marks.

The working does not have to be repeated if seen in part (b)(i).

3

(c) Resistance forces <u>vary with speed</u> (or velocity)
OR Speed (or velocity) changes (or increases)
OR It accelerates

B1: Correct explanation. Must not mention friction in main argument

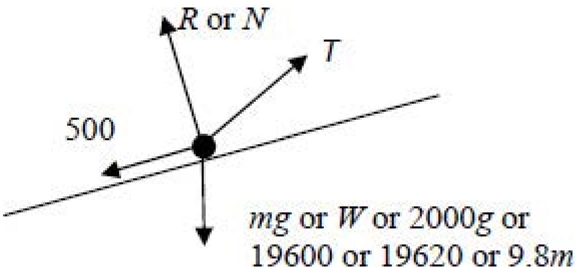
B1

1

[13]

Q32.

(a)



B1: R, 500 and mg correct

B1

B1: Tension in roughly correct direction.

If more than four forces shown, do not award more than one mark.

Note all forces must be shown as arrows and have labels.

Note some candidates may draw the force diagram in the section with the question.

Components can be ignored if shown in a different notation eg dashed arrows.

B1

2

(b) 2000 0.6 = T cos12° – 500 – 2000 9.8sin5°

M1: Resolving parallel to the slope to obtain a four term equation of motion. The weight and tension terms must be resolved.

A1: Correct terms.

A1: Correct signs.

M1A1A1

$T = \dfrac{1200 + 500 + 19600sin5°}{cos12°}$

$\left(= \dfrac{3408.25}{cos12°}\right)$

(= 3484.4)

dM1: Solving for T.

dM1

= 3480 (to 3sf) AG

A1: Correct tension. AWRT 3480.

Allow AWRT 3490 from use of g = 9.81.

A1

5

[7]

Q33.

(a) $\mathbf{r} = \int v\,dt = (4t + t^3)\mathbf{i} + (12t - 4t^2)\mathbf{j} + \mathbf{c}$

*M1: either **i** or **j** term correct.*
*Condone no **c***

M1A1

When $t = 0$, $\mathbf{r} = 5\mathbf{i} - 7\mathbf{j}$
$\mathbf{c} = 5\mathbf{i} - 7\mathbf{j}$

*Any attempt at **c***

M1

$\mathbf{r} = (5 + 4t + t^3)\mathbf{i} + (-7 + 12t - 4t^2)\mathbf{j}$

A1

4

(b) $\mathbf{a} = \frac{dv}{dt}$

$\mathbf{a} = 6t\,\mathbf{i} - 8\,\mathbf{j}$

M1: either term correct

M1A1

2

(c) Using $\mathbf{F} = m\mathbf{a}$

Or: using $\mathbf{F} = m\mathbf{a}$

M1

$\mathbf{F} = 2(6t\,\mathbf{i} - 8\,\mathbf{j})$

$\mathbf{F} = 2(6t\,\mathbf{i} - 8\,\mathbf{j})$

$= 12t\,\mathbf{i} - 16\,\mathbf{j}$

When $t = 1$, $\mathbf{F} = 12\,\mathbf{i} - 16\,\mathbf{j}$

A1

$\therefore$ Magnitude of force is $(144t^2 + 256)^{\frac{1}{2}}$ when $t = 1$

Magnitude of force is $(144 + 256)^{\frac{1}{2}}$

M1

= 20 N

= 20 N

A1 4

[10]

Q34.

(a) (i) $F = 65g - 260v$

Accept $260v - 65g$

$= 65(9.8 - 4v)$

AG must see $65v$ *or 260*

B1 1

(ii) Using $F = ma$

$65\frac{dv}{dt} = 65(9.8 - 4v)$

Need to see terms in m *(condone – sign)*

M1

$\frac{dv}{dt} = -4(v - 2.45)$

AG

A1 2

(b) $\frac{1}{v - 2.45}\frac{dv}{dt} = -4$

B1

$\int \frac{1}{v - 2.45}\,dv = -\int 4\,dt$

$\ln(v - 2.45) = -4t + c$

M1: log side correct

M1

$-4t + c$

A1

$v - 2.45 = Ce^{-4t}$

$t = 0$, $v = 19.6$

$\therefore C = 17.15$ or $e^{2.84}$

Or $c = \ln 17.15$ or 2.84

A1

$\therefore v = 2.45 + 17.15e^{-4t}$ $2.45 + 17.2e^{-4t}$

A1

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Q35.

(a) (i) $0.6^2 = 0^2 + 2a \times 0.9$

M1: Correct use of constant acceleration equation with u = 0 to find a.

A1: Correct equation.

M1A1

$a = \frac{0.6^2}{1.8} = 0.2 \text{ ms}^{-2}$ **AG**

A1: Correct a but some intermediate working must be seen.

A1

Note that $0^2 = 0.6^2 + 2a \times 0.9$

Scores M0A0A0

Verification methods require a conclusion for full marks to be awarded.

Condone seeing just the second line of working.

3

(ii) $0.9 = \frac{1}{2}(0 + 0.6)t$

M1: Correct use of constant acceleration

equation
with u = 0 (and a = 0.2 if needed) to find t.

M1

$t = \frac{0.9}{0.3}$ = 3 seconds

A1: Correct time.

A1

Note: Do not penalise $0.9 = \frac{1}{2}(0.6 + 0)t$ in the first method.

OR

$0.6 = 0 + 0.2t$

(M1)

$t = \frac{0.6}{0.2}$ = 3 seconds

(A1)

Note: $0 = 0.6 + 0.2t$ scores M0A0 in the second method.

OR

$0.9 = \frac{1}{2} 0.2t^2$

(M1)

t = 3 seconds

(A1)

2

(b) $T - 800 \times 9.8 = 800 \times 0.2$

M1: Three term equation of motion. Must have these three terms but can have incorrect signs. Must use a = 0.2

A1: Correct equation with correct signs.

(Allow 800g)

M1A1

T = 7840 + 160 = 8000 N

A1: Correct tension.

A1

Accept 8008 or 8010 from use of g = 9.81.

3

[8]